Transcript Intelligence — Findings & Pipeline

Take-home assignment: 100 transcripts processed through a hybrid pipeline to surface three insights for stakeholders.

3 insights: churn risk concentration · communication gap · convergent feature gaps

Pipeline: 6 stages, hybrid (rules + embeddings + selective LLM), ~$0.30 in API costs

Validation: matches pre-computed labels on 52 of 61 churn signals, finds 7 additional

Context: the March 2026 Detect Outage

In mid-March 2026, Aegis Detect's event-processing pipeline had a six-hour cascading failure. Zero threat visibility for affected customers. The technical fix shipped in 30 days.

The business damage didn't.

- 8 customers entered *active churn-risk territory* within 10 days
- 39 of 100 calls show customers frustrated by *silence*, not by the bug
- 5 product gaps were independently raised by *both customers and engineers* — including the missing pipeline-health monitoring that could have prevented the outage

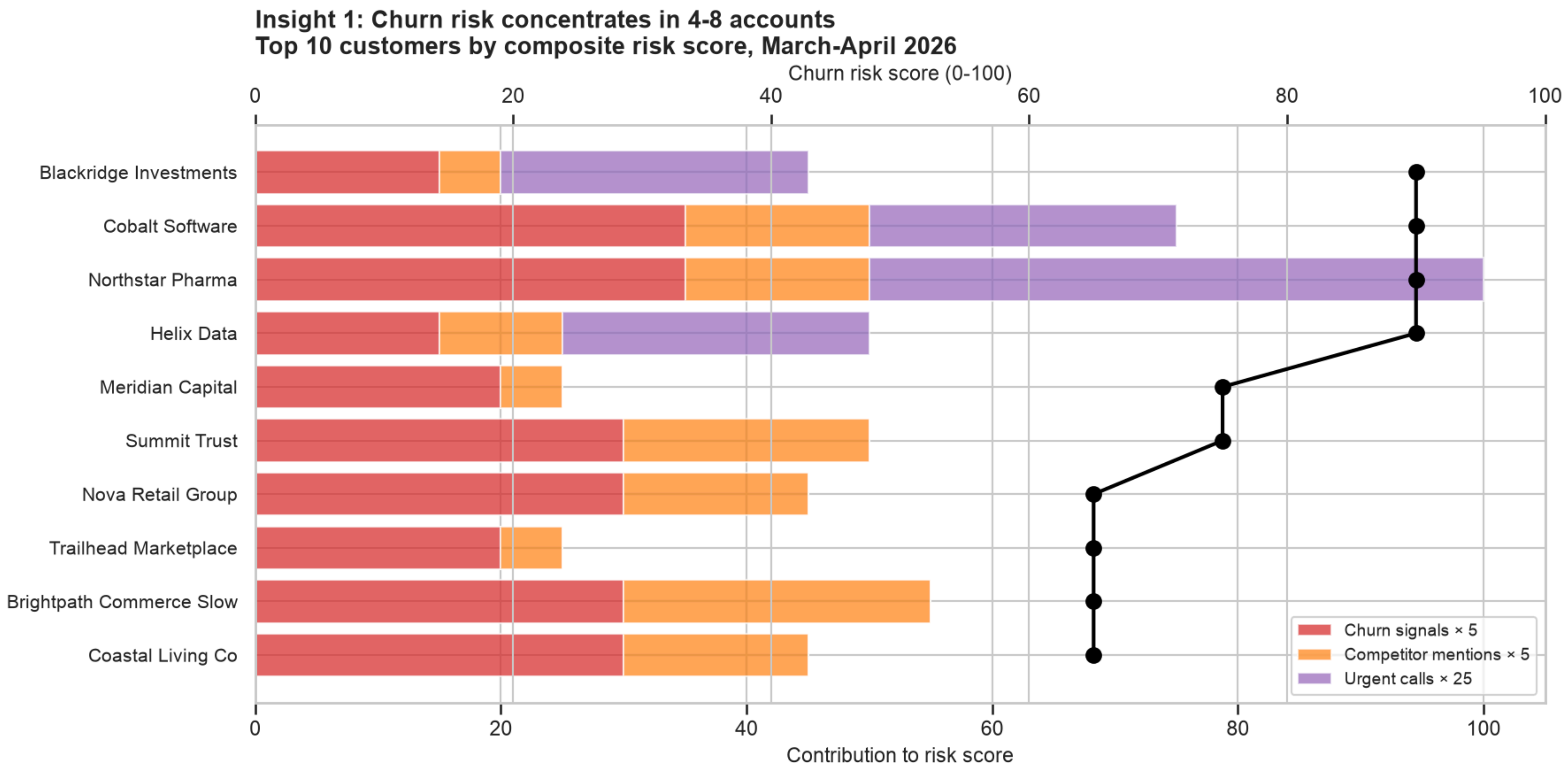
A tool that surfaces these signals in real time, per stakeholder, would have flagged 4 accounts weeks before they escalated.

Insight 1 — Churn risk concentrates in 4-8 accounts

After the outage, 14 accounts scored HIGH churn risk; 4-8 of them account for the bulk of competitor mentions and churn signals.

- **30 of 100 calls** name a competitor by name (SentinelShield, CyberNova, VaultEdge)
- **Same customers repeat**: Blackridge, Cobalt, Northstar, Helix, Meridian, Summit Trust
- **Internal Apr 24 Win/Loss call**: 34 closed-lost deals totaling $2.1M ACV in Q1, with SentinelShield as primary winner

A real-time churn risk score would have flagged Blackridge, Cobalt, and Northstar 30 days before they escalated.



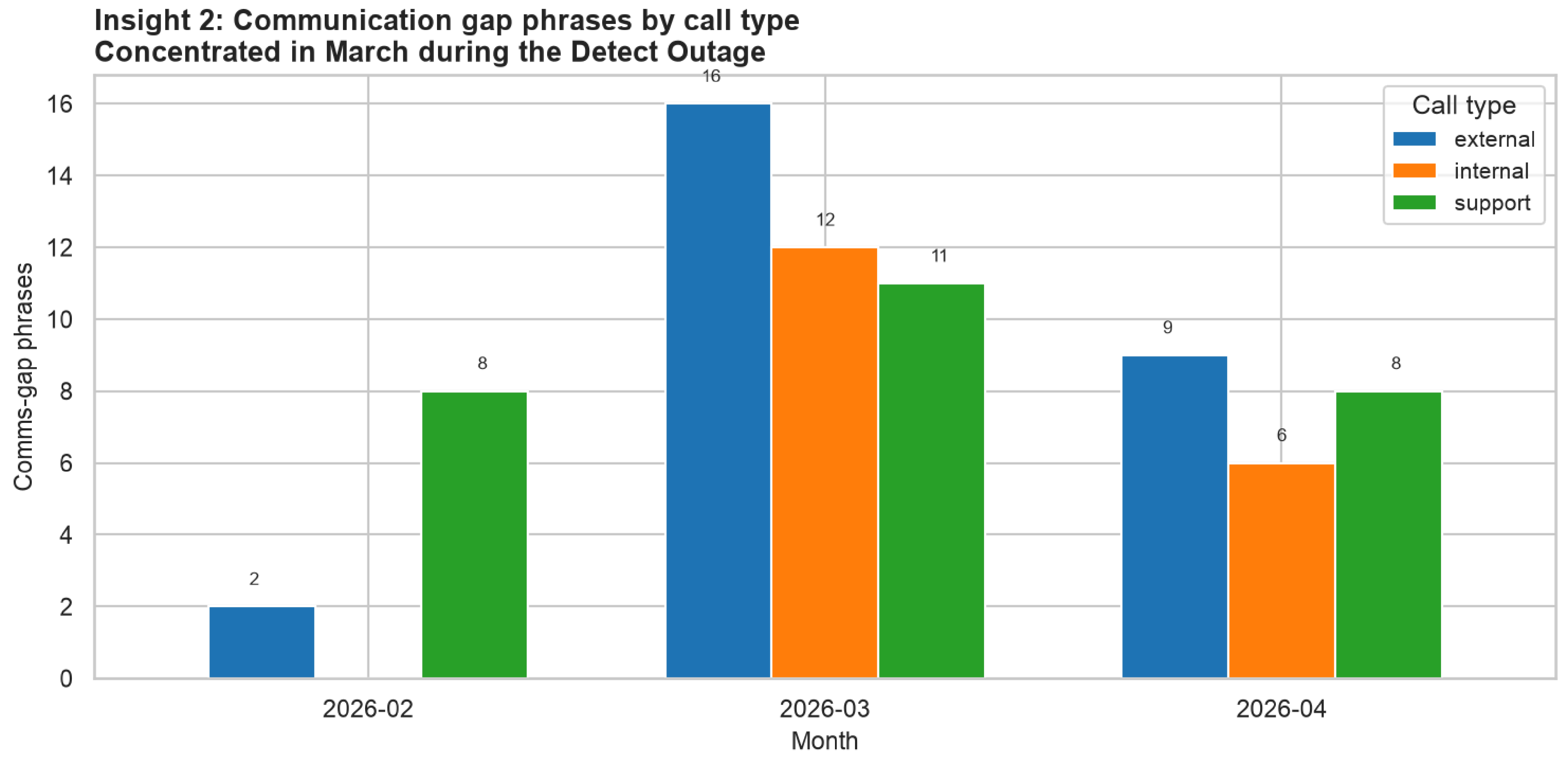
Insight 1: Churn risk concentrates in 4-8 accounts
Top 10 customers by composite risk score, March-April 2026
Churn risk score (0-100)
0
20
40
60
80
100
Blackridge Investments
Cobalt Software
Northstar Pharma
Helix Data
Meridian Capital
Summit Trust
Nova Retail Group
Trailhead Marketplace
Brightpath Commerce Slow
Coastal Living Co
Churn signals × 5
Competitor mentions × 5
Urgent calls × 25
Contribution to risk score

Insight 2 — The wound is the silence, not the bug

39 of 100 calls (59 total mentions) contain communication-gap language. Concentrated in March during the outage window.

- "We didn't get any notification from Aegis" — Paula Schneider, Ridgeline
- "A customer had to report the outage before Aegis detected it internally" — Lauren Bishop, Cobalt
- "Customers feel like they're flying blind with Detect" — Diana Reeves, internal post-mortem

The process fix may matter more than the engineering fix. One proactive-comms trigger could prevent the next outage from doing the same brand damage.



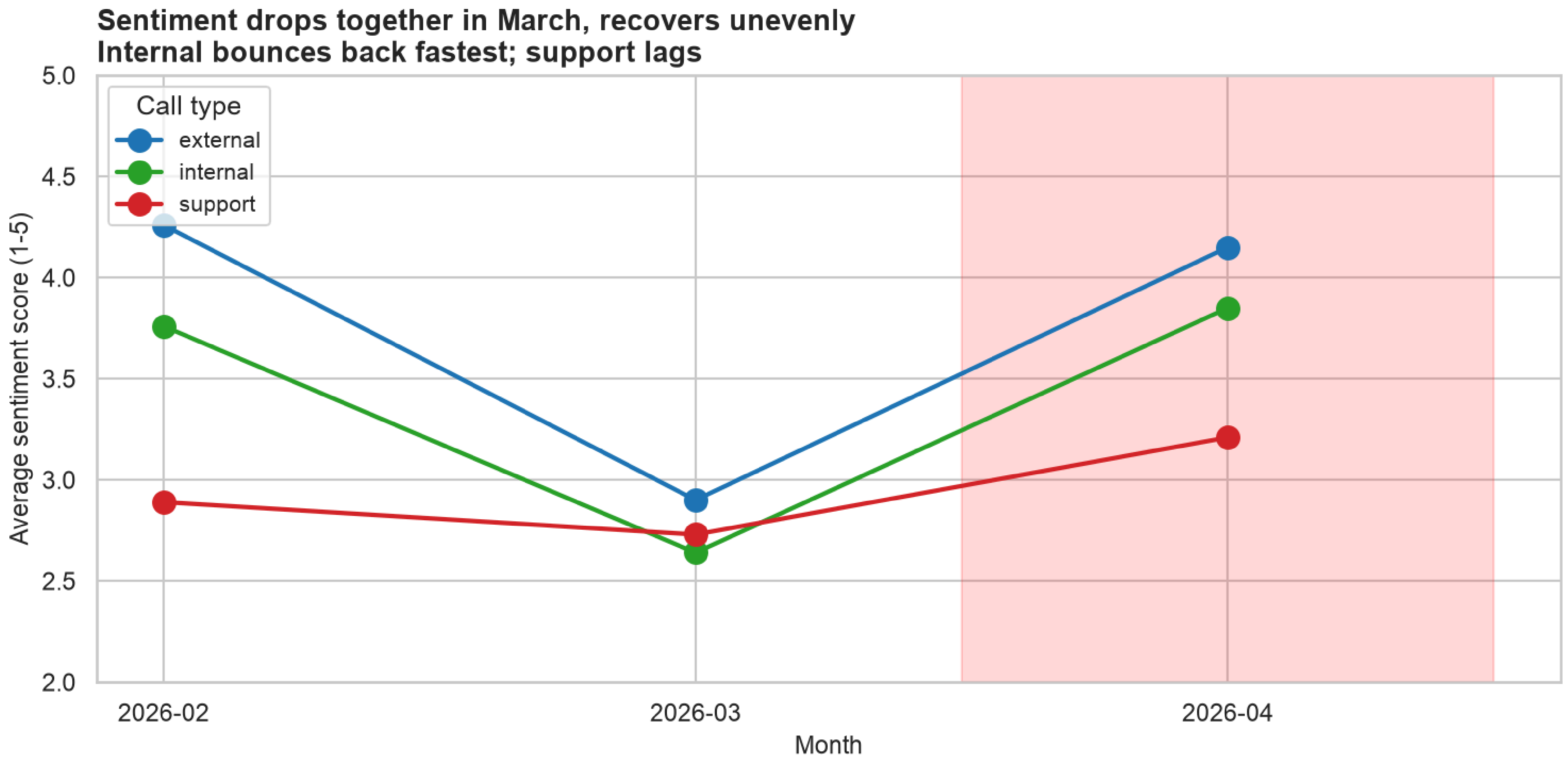
Insight 2: Communication gap phrases by call type
Concentrated in March during the Detect Outage
Call type
external
internal
support
Comms-gap phrases
Month
2026-02
2026-03
2026-04
0
2
4
6
8
10
12
14
16
2
8
16
12
11
9
6
8

Sentiment recovery is uneven

All three call types dropped in March. Recovery differs by audience:

- **Internal**: 2.64 → 3.85 — bounced back above pre-outage levels
- **External**: 2.90 → 4.15 — back to ~normal
- **Support**: 2.73 → 3.21 — *still below* February baseline

Customers who actually had a problem didn't fully regain trust. Closing tickets ≠ closing the trust gap.


Sentiment drops together in March, recovers unevenly
Internal bounces back fastest; support lags
Call type
external
internal
support
Average sentiment score (1-5)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
2026-02
2026-03
2026-04
Month

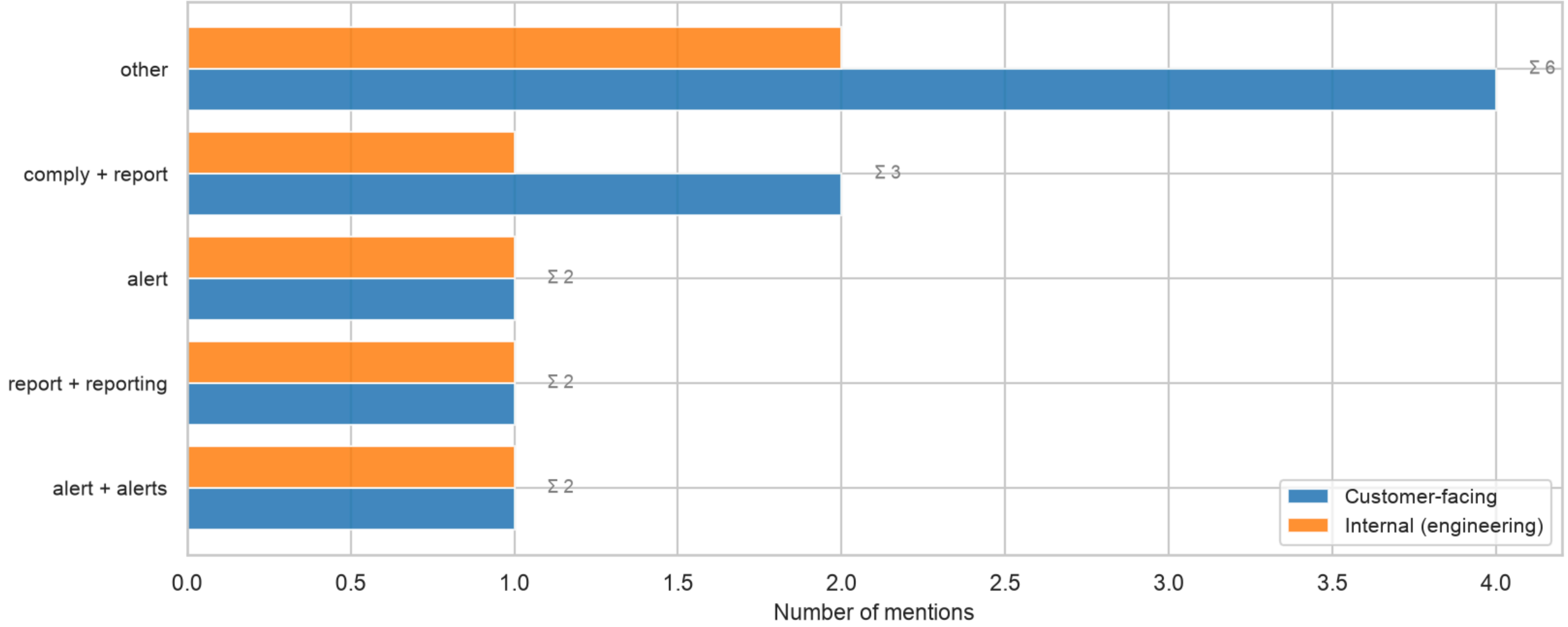
Insight 3 — Convergent feature gaps

5 product gaps were independently identified by *both* customers and engineers.

- **Pipeline health visibility** — customer (Pinnacle, Apr 16) + engineer (Ravi, Apr 28 retro)
- **Heartbeat alerting** — engineer (Megan, Mar 18 post-mortem) + customer (Blackridge, Apr 21)
- **SSO / MFA refactor** — customer (Frostbyte, Apr 25) + engineer (Tyler, Mar 28 retro)
- **Comply v2 launch scope** — multiple customers + engineer (Megan, Mar 24 launch readiness)
- **Restore UX** — customer (Meridian, Mar 26) + engineer (Sofia, Feb 23 planning)

Convergent gaps are pre-validated roadmap priorities — customer demand + engineering awareness in the same window.

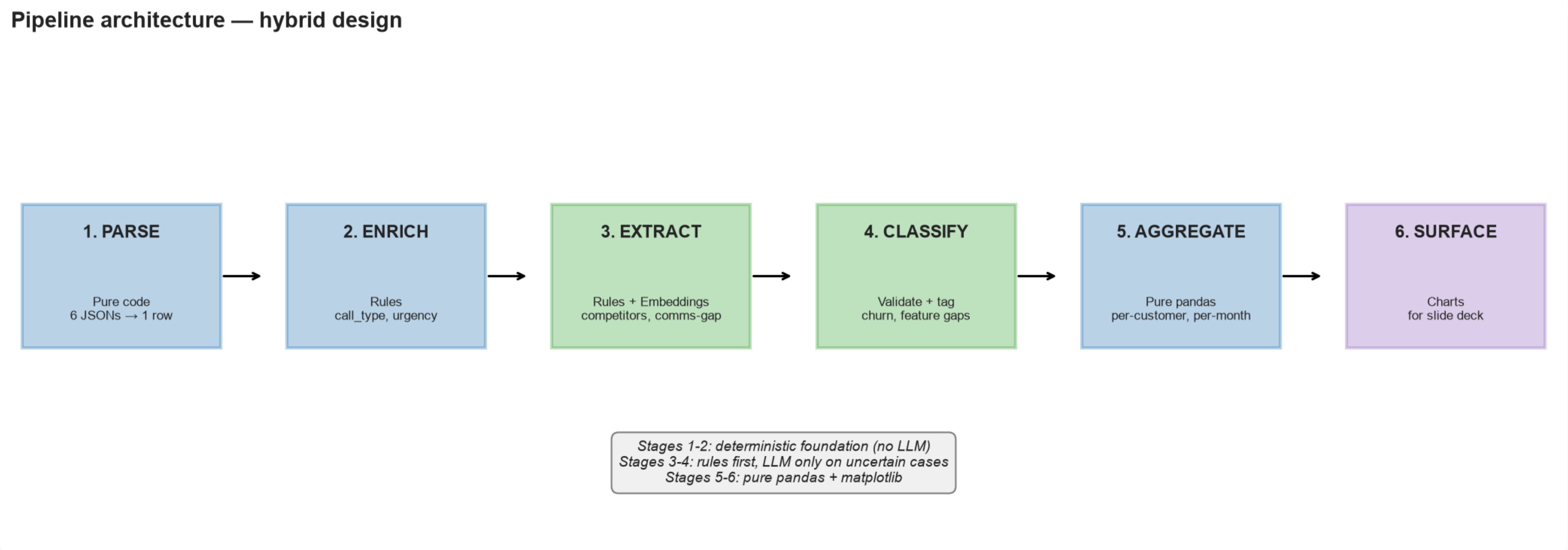
Insight 3: Convergent feature gaps
Same product gap raised independently by customers and engineers



Pipeline architecture

Hybrid design — rules first, embeddings for clustering, LLM only on uncertain cases.

- **Stages 1-2**: deterministic foundation, no LLM (~$0 cost)
- **Stages 3-4**: rules extract ~80% of signals at near-100% precision; LLM fires only on uncertain cases (~$0.001-0.005 per call)
- **Stages 5-6**: pure pandas + matplotlib, fully auditable business logic



Validation — agreement with pre-computed labels

The dataset includes pre-computed `keyMoments` in `summary.json`. We treat these as *reference labels*, not ground truth, and compare our pipeline's output to them.

Metric	Our extraction	Pre-computed	Agreement
Calls with churn_signal	59	61	52 (both agree) · 7 (we found more) · 0 (we missed)
Feature gaps	51 (36 customer-facing + 15 internal)	51	100% recall, convergent view identifies 5 cross-side gaps
Competitor mentions	98 mentions across 30 calls	n/a	New signal — not in pre-computed
Comms-gap phrases	57 across 38 calls	n/a	New signal — not in pre-computed

Where we disagree with pre-computed labels, those are the most interesting findings *— usually cases where the source vendor missed a signal we caught via rules + competitor mentions.*

Cost & scale story

Pure-LLM-only analysis is fine at 100 transcripts. It doesn't scale.

Volume	Pure LLM	Hybrid (this pipeline)
100 calls (this assignment)	~$0.50	**~$0.30**
50,000 calls / year	$250	**$25**
500,000 calls / year	$2,500	**$250**
5,000,000 calls / year	$25,000	**$2,500**

Plus: **auditability** (every signal traceable to a rule or quote), **reproducibility** (deterministic for non-LLM stages), and **latency** (<1s for 90% of calls).

What we'd build next

1. **Real-time churn-risk dashboard** for CS leaders, fed by this pipeline. Top-10 at-risk accounts updated daily, with linked transcripts and quotes for each signal.
2. **Proactive-comms trigger**: when an incident is detected, auto-surface the customer-affected list within 30 minutes. Closes the comms-gap before it metastasizes.
3. **Convergent-gap radar**: a weekly digest for the Head of Product, showing gaps with both customer demand and engineering awareness. Should be the top of every roadmap conversation.
4. **Account-health composite**: per-customer score combining sentiment trend, churn signals, action-item closure rate, and comms-gap mentions. QBR-ready.

Q&A

Common questions and how to answer them are in `INTERVIEW_PREP.md` in this repo.

- "Why hybrid instead of pure LLM?" — cost, audit, latency (see Q&A doc)
- "How would you scale to 1M transcripts?" — same architecture, parallelize + DB
- "What did the pre-computed labels get wrong?" — 7 cases where our extraction found more
- "What's the most surprising finding?" — the convergent gaps (same product, two voices)

All code in `pipeline/`, all charts in `outputs/charts/`, all numbers traceable to `data/processed/`.